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University of the Punjab, Lahore

Smart Academic Communication Platform (SACP)

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Statement of Submission

In this regard, we formally declare that the project titled "**Smart Academic Communication Platform**", which has been done in fulfillment of the requirements for the **Bachelor of Science in Computer Science (BSCS)** degree at **Faculty of Computing & Information Technology, University of the Punjab, Lahore**, is our original effort.

This project has been conducted under the supervision of the project supervisor assigned to us. It is hereby declared that this project has never been submitted, wholly or partially, for obtaining any degree or diploma from any other university or institution before.

We have properly mentioned all the references used in this project.

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Special Thanks

- Person A
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Abstract

Academic institutions often encounter difficulties in organizing effective communication between administrators, faculty members, and students. Traditional communication systems are manual, inefficient, delay-prone, and hard to scale. Important notifications, such as warnings about attendance, assignment deadlines, test schedules, and institutional communications, may be late and not delivered to their respective recipients on time. This reduces the effectiveness of the communication process and causes organizational problems.

Smart Academic Communication Platform is an integrated, intelligence, automated, and web-based tool to facilitate efficient academic communications. It offers rule-based automation of institutional communication processes and personalization of messages based on institutional data to create context-aware and personalized messages. Such features include dynamic messages templates, campaigns scheduling, target audience selection, notification tracking, and analytics.

The architecture of the project relies on a distributed 3-tier structure that utilizes React for front-end applications, ASP.NET core backend services, Node.JS-based real-time communications, and AI implemented services using Python. Machine learning algorithms can help to classify and prioritize emails, thus preventing spam issues and focusing more on relevant academic messages. Other functionalities offered by SACP include real-time chat options, communication logging tools, reports, and documents sharing.

The main purpose of the project is to provide institutional communication solutions with automation capabilities and improved message delivery, which allows for reduced manual efforts and intelligent assistance with communication tasks.

The expected result of the project development is a functional academic communication platform that provides scalable, usable, centralized management and analytics capabilities.

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Chapter 1

INTRODUCTION

1.1 Background

Communication plays a critical role in ensuring that higher education institutions run smoothly. There are constant interactions between administrators, lecturers, and learners with regard to lectures, exams, assignment deadlines, scheduling, notifications, and other academics. Most universities and colleges use conventional communication means such as emails, social media groups, printed notices, or external messaging apps in their interactions. The problem with these communication strategies is that they lead to ineffective messaging, poor message management, lack of personalization, and difficult large-scale communications.

In light of the current digitized world and data-driven systems, institutions need communication platforms which can automatically streamline their communication procedures. There are existing solutions which are learning management-oriented or notification-oriented but they lack communication management, automation of communication workflows, analytics, intelligent prioritization, and targeting features.

Smart Academic Communication Platform (SACP) will provide an innovative, central, and intelligent communication solution specifically designed to serve the needs of institutions. Automation will be achieved using rule-based workflows, campaigns, dynamic templates, real-time messaging, and email classification and prioritization using artificial intelligence technology. Integration with academic records will ensure that there is personalized and context-aware communication among administrators, educators, and learners.

Distributed architecture principles will also be addressed through the development of specific layers for frontend, backend core services, real-time communication services, and artificial intelligence services. It will provide an automated, analytical, and AI-powered system which aims to streamline academic institution communication procedures.

1.2 Problem Statement

In academic institutions, many communication issues arise due to the use of manual and semi-manual mechanisms. Academic notifications like warnings about poor attendance, deadlines for assignments, exam schedules, fee collection reminders, and other important announcements are often made through inefficient or inconsistent communication processes. The current system of communication is ineffective because it does not offer personalization, automation, centralization, and intelligent message priority.

In addition, organizations operating thousands of students and faculty face the following challenges:

- Sending targeted and personalized communications
- Managing large communication campaigns
- Monitoring deliverability and engagement metrics
- Filtering out spam and irrelevant content
- Automating repetitive communication processes
- Centralizing communications record-keeping and analysis

- Securing role-based communication lines

Manual systems fail to provide suitable automation capabilities and real-time collaboration opportunities. With the increase in communication volume, organizations need scalable technology solutions to automate and optimize their communication processes.

There is thus a need for a centralized, automated, and intelligence-based system to support personalized communications, workflow automation, communication analytics, real-time communications, and machine-learning-based message priorities.

1.3 Objectives

The main objective of the SACP solution is to create a smart communication platform for academic institutions, which will automate the process of communication and increase its efficiency.

The key objectives of the project include the following:

- Create a centralized platform for managing academic communication.
- Utilize dynamic templates to support personalized communication.
- Introduce automated and rule-based workflows to be activated as a result of certain academic events, such as missed class time, due dates for assignments, exam dates, etc.
- Manage campaigns and send bulk emails/messages.
- Allow for targeted communication according to academic data.
- Keep track of emails via logs and delivery reports.
- Develop analytics dashboards to monitor communication performance.
- Apply machine learning techniques to categorize and prioritize emails to ensure that only important emails are delivered.
- Introduce safe and instant communication capabilities within the platform.
- Decrease the manual work required to send emails and increase efficiency.

1.4 Scope of the Project

The scope of the Smart Academic Communication Platform (SACP) includes design and implementation of the web-based, distributed communication platform specifically for academic institutions. The goal of the platform is to improve the institutional communication capabilities through automation, personalization, analytics, and smart communication management.

The key functional domains of the platform include:

- User authentication & role-based authorization
- Management of academic records
- Template management (dynamic templates)

- Campaign creation & scheduling
- Rule based workflow automation
- Email & notifications service
- Real-time one-to-one communication
- AI-powered email classification/prioritization
- Communication analytics & reporting
- Document/file exchange

Key target audiences of the SACP platform are the following:

- Academic Administrators
- Faculty members
- Students

Regarding architecture of the application, the platform follows the distributed approach, where frontend services, backend/core services, real-time communications services, and AI-based services are independently developed to increase maintainability and scale of the platform.

Among others, exclusion domains of the project include the following:

- Payment and financial operations
- Complete functionality of LMS platform
- Social messenger type applications like WhatsApp, Messenger

It is worth noting that the core of the project does not include any course management functions.

1.5 Methodology Summary

The methodology of development on which the SACP has been developed uses an iterative approach for software development. The software engineering methodology also includes the integration of other concepts such as the concept of distributed architecture, databases, and use of machine learning techniques.

The software development process includes the following stages:

1.5.1 Requirement Analysis

We analyze communication related problems within academic institutions in this phase. Based on the needs of academic institutions functional and non-functional requirements are identified.

1.5.2 System Design

In this phase, we design the entire system of SACP including system architecture, database design, workflow models, communication modules, automation engine, and AI integration components are designed using UML diagrams, ERDs, and architectural models

1.5.3 Frontend Development

In this phase, we design and develop user interfaces using React and modern UI libraries to provide user-friendly and responsive interfaces for administrators, faculty members, and students.

1.5.4 Backend Development

In this phase, all services related to backend core are implemented using ASP.NET Core Web APIs. These services handle authentication & authorization, campaign & template management, automation workflows, analytics and AI orchestration with Python services.

We'll also develop real-time backend services for communication module and real time analytics for dashboard using SignalR and Redis within .NET ecosystem.

1.5.5 AI/ML Service Development

In this phase, we'll develop AI/ML services for email classification and prioritization using Python. We'll develop, finetune, or create machine learning models and NLP techniques to achieve this.

1.5.6 Database Integration

In this phase, we'll setup either PostgreSQL or Azure SQL database to store academic records, communication logs, templates, campaigns and analytics data. We'll use a centralized database to avoid duplication, redundancy, and confusions.

1.5.7 Testing and Validation

In this phase, we'll conduct unit testing, integration testing, API testing, AI model evaluation, security testing, and performance testing techniques to ensure overall reliability and functionality of the SACP.

1.5.8 Deployment and Evaluation

In this phase, we'll deploy the final system in a distributed environment considering scalability, usability, and communication performance among system components.

1.6 Tools and Technologies Used

The Smart Academic Communication Platform (SACP) is developed with modern web, backend, database and AI technologies to provide scalability, maintainability and performance.

1.6.1 Frontend Technologies

- React
- HTML5
- CSS3
- JavaScript
- Tailwind CSS / Material UI
- Redux
- Charts.js / Recharts

1.6.2 Backend Technologies

- C#
- ASP.NET Core Web APIs
- Entity Framework Core
- LINQ
- JWT Authentication
- SignalR
- Redis
- Hangfire / Quartz.NET

1.6.3 AI/ML Services

- Python
- FastAPI / Flask
- Scikit-Learn
- Pandas
- NumPy
- spaCy / NLTK
- Matplotlib

1.6.4 Database System

- PostgreSQL / Azure SQL

1.6.5 Development and Collaboration Tools

- Visual Studio
- Visual Studio Code
- PyCharm
- Git & GitHub
- Swagger / Postman
- Draw.io
- PlantUML
- Figma
- Jira / Trello
- MS Office

1.6.7 Deployment Environment

- Linux
- Cloud-based deployment support
- Distributed service architecture

1.7 Summary

This chapter introduced the Smart Academic Communication Platform (SACP), its background, aims, scope, methodology and technologies used. The chapter explored the communication challenges that academic institutions face and explained how automation, distributed architecture, and machine learning can help improve communication workflows. The next chapter includes the literature review and analysis of existing systems related to academic communication, notification management, collaboration platforms, and AI-based communication solutions.

Chapter 2

LITERATURE REVIEW

2.1 Introduction

Academic institutions mostly rely on digital communication platforms to manage their interactions among faculty members and students. A simple, effective, and autonomous communication is essential for delivering timely messages, but still many institutions use tools which are made for general public and specialized for their specific use cases like tracking engagement, sending messages to large audience on specific times or events, etc.

There are many modern platforms, such as Slack, Microsoft Teams, Discord, etc., which provide real-time communication and collaboration. But email classification and campaign systems, template systems, and automated workflows for specific needs of academic institutions are missing. Current solutions typically focus on isolated communication tasks and rarely offer a unified academic communication system for their specific needs.

The proposed Smart Academic Communication Platform (SACP) seeks to overcome these challenges by integrating all necessary features very specific to academic institutions into a single distributed system architecture.

This chapter reviews existing communication platforms, notification systems, automation tools, and intelligent messaging solutions related to SACP.

2.2 Summary of Existing Systems and Research

2.2.1 Slack

Feature Category	***Slack Capabilities***	***What Slack Lacks***	***Proposed SACP Enhancements***
Core Communication	It offers real-time & threaded messaging, dedicated classroom/announcement channels, and audio/video meetings & file sharing	It lacks large-scale SMS/email broadcasting and automated academic notifications	It offers rule-based academic automation and workflows for personalized notifications
Integrations	It integrates with 3rd-party apps & productivity tools like LMS integration (Canvas/Blackboard) via LTI Course Bots	It lacks centralized campaign management	It offers scheduled communication campaigns and unified location for multiple academic record sources
Intelligence & Analytics	It offers basic workspace synchronization	It lacks mL-based email classification and advanced academic analytics	It offers AI-based message prioritization and comprehensive communication & workflow dashboard

Table 1: Slack Gap

2.2.2 Microsoft Teams

Feature Category	*Microsoft Teams Capabilities*	*What Teams Lacks*	*Proposed SACP Enhancements*
Core Communication	It offers real-time chat & class channels, video conferencing & calendar sync, and native assignment/classwork management	It lacks native external messaging (SMS/WhatsApp), and multi-layered distributed structure	It offers multi-layer distributed architecture for external messaging, and optional multi-channel outreach
Integrations	It offers deep Microsoft Office app ecosystem, and LMS sync & School Data Sync (SDS) for grades	It lacks flexible, institutional campaign management tools	It offers centralized communication campaigns, and academic event-driven automation
Intelligence & Analytics	It offers basic classwork activity tracking	It lacks AI-driven priority assignment, and advanced notification analytics	It offers intelligent email filtering, and analytics-driven communication workflows

Table 2: Microsoft Teams Gap

2.2.3 Discord

Feature Category	*Discord Capabilities*	*What Discord Lacks*	*Proposed SACP Enhancements*
Core Communication	It offers real-time text, voice, & video chat, community-based servers & channels, and screen sharing & basic file sharing	It lacks out-of-platform notifications, and centralized communication tracking	It offers centralized academic communications tracking, and AI-driven communications prioritization
Integrations	It offers multi-platform accessibility, and basic announcement channels	It lacks native LMS integrations, and centralized academic record syncing	It offers seamless integration of academic records, and workflow automation built for institutions
Administration	If offers informal server roles and basic text permissions	It lacks proper institutional user structuring, and academic analytics & reporting	It offers role-based academic communication management, and admin dashboard with academic & operational analytics

Table 3: Discord Gap

2.2.4 Rocket.CHAT

Feature Category	*Rocket.Chat Capabilities*	*What Rocket.Chat Lacks*	*Proposed SACP Enhancements*
Core Communication	It offers real-time chat & channel collaboration, secure, private messaging, and direct file sharing capabilities	It lacks academic workflow tools, and centralized, campus-wide logging	It offers built-in academic communication workflows, and centralized communication logging
Integrations	It offers flexible third-party application integrations	If lacks development effort required to adapt for complex academic logic	It offers academic event-triggered automation, and native, plug-and-play institutional triggers
Security & Analytics	It offers on-premises, self-hosted data control, and compliance-ready secure environments	It lacks intelligent message classification or filtering, and dedicated institutional analytics dashboards	It offers AI-based communication classification, and comprehensive analytics and reporting modules

Table 4: Rocket.CHAT Gap

2.2.5 MailChimp

Feature Category	*MailChimp Capabilities*	*What MailChimp Lacks*	*Proposed SACP Enhancements*
Core Communication	It offers bulk campaign creation & scheduling, drag-and-drop template libraries, and dynamic placeholder fields	It lacks native multi-channel academic networks, and institutional role mapping	It offers campaign management module, and template management system natively with academic role distribution.
Audience & Automation	It offers audience segmentation based on lists, and standard user event-triggered emails	It lacks native integration with academic registers, and event engines built for school systems	It natively offers academic data management, and automation rule engine to make communication automated and context-aware with engagement analytics

Table 5: MailChimp Gap

2.2.6 SendGrid (Twillio SendGrid)

Feature Category	*SendGrid Capabilities*	*What SendGrid Lacks*	*Proposed SACP Enhancements*
Messaging Infrastructure	It offers high-volume transactional email delivery, and API-driven framework optimized for heavy messaging loads	It lacks built-in application UI for non-technical academic staff	It offers core messaging service with ASP.NET Core & Hangfire backend optimizations
Tracking & Logging	It offers real-time processing receipts, and detailed delivery logs & bounce-rate tracking	It lacks academic context tracking (linking logs directly to student/faculty profiles)	It offers detailed communication logs, and centralized student-to-message history mapping
Architecture & Performance	It contains heavy infrastructure designed for offloading primary app threads	It lacks pre-built educational logic, automation rules, or template engines	It contains Hangfire-driven background processing integration, and native academic campaign & rule engines

Table 6: SendGrid Gap

2.2.7 Braze

Feature Category	*Braze Capabilities*	*What Braze Lacks*	*Proposed SACP Enhancements*
Automation & Journeys	It offers real-time lifecycle behavior triggers, and visual cross-channel journey building (Canvas Flow)	It lacks native academic-specific entity triggers (e.g., GPA drops, prerequisite failures)	It offers automation rule engine, and pre-configured institutional event templates
Data & Integrations	It offers streaming data ingestion from warehouses, and dynamic content personalization via external APIs	It lacks integrations with standard school registers and educational records	It offers academic data management to store all academic records
Intelligence & Analytics	It offers real-time user interaction tracking, and predictive churn and behavioral AI modeling	It lacks contextual academic classification (e.g., urgent administrative alerts vs. student club notifications)	It offers analytics and reporting. and AI-based academic communication classification

Table 7: Braze Gap

2.2.8 Moodle

Feature Category	*Moodle LMS Capabilities*	*What Moodle Lacks*	*Proposed SACP Enhancements*
Core Communication	It offers native direct 1-to-1 & group chat, in-platform web alerts & bulk messaging, and mobile push via AirNotifier (APNs/FCM)	It lacks automation, and direct commercial multi-channel fallback	It offers built-in academic event-triggered alerts
Campaigns & Data	It offers simple bulk student messaging by course, and custom message API for plugins	It lacks centralized campaign scheduling interfaces, and reusable template engines with record data	It offers campaign scheduling & management UI, dynamic message templates with record variables
Automation & Analytics	It offers per-user delivery channel preferences, and administrator institution-wide defaults	It lacks complex, rule-based academic trigger engines, open, read, and bounce-rate tracking, and intelligent message triage and sorting	It offers rule-based workflow automation engine, delivery Tracking & Analytics dashboard, and AI/ML Email Classification & Prioritization

Table 8: Moodle Gap

2.2.9 WhatsApp

Feature Category	*WhatsApp Capabilities*	*What WhatsApp Lacks*	*Proposed SACP Enhancements*
Core Communication	It offers instant, high-engagement text & media delivery, interactive messaging components, and end-to-end encrypted delivery channels	It lacks unified, cross-platform email fallbacks, and flexible, unvetted outbound text broadcasting	It offers centralized multi-channel management layer, and unified email, SMS, and messaging orchestration
Governance & Data	It offers strict opt-in handling & privacy protection, and commercial template pre-approval systems	It lacks direct out-of-the-box integration with student information systems	It offers direct integration with institutional academic records, and centralized academic communication governance
Automation & Intelligence	It offers basic interactive messaging automated replies in its business version.	It lacks triggers tied to real-time school events, advanced notification performance reporting, and ML-based message triage and prioritization	It offers academic event-triggered automation engine, AI/ML message classification & prioritization, and built-in campaign management & delivery analytics

Table 9: WhatsApp Gap

2.2.10 Spark Mail

Feature Category	*Spark Mail Capabilities*	*What Spark Mail Lacks*	*Proposed SACP Enhancements*
Core Communication	It offers smart Inbox filtering (Personal, Notifications, News), real-time team email sharing & commenting, and unified client for standard email ecosystems	It lacks centralized multi-channel broadcasting (SMS/WhatsApp), and native cross-platform conversational chat layers	It offers centralized multi-channel orchestration layer, and ntegrated real-time conversational chat module
AI Capabilities	It offers Spark +AI writing & drafting assistance, and automated thread summarization & quick replies, and basic priority sorting for standard emails	It lacks classification based on academic context (e.g., grading alerts vs. campus events)	It offers AI-based academic communication classification, and ML-driven institutional prioritization
Automation & Analytics	It offers basic personal email templates and snoozing	It lacks triggers tied to external databases or school events, campus-wide delivery tracking and audit logs, and centralized administrator monitoring dashboards	It offers academic event-triggered automation engine, campaign administration module, and institutional analytics & reporting

Table 10: Spark Mail Gap

2.2.11 Microsoft Outlook

Feature Category	*Outlook AI Capabilities*	*What Outlook AI Lacks*	*Proposed SACP Enhancements*
Core Communication	It offers deep Microsoft 365 ecosystem fluid sync, advanced Mail Merge for mass personalization, and real-time meeting readiness dashboards	It lacks non-email channels (SMS/WhatsApp), and native student conversational chat partitions	It offers multi-channel broadcasting with integrated real-time conversational chat module
AI Capabilities	It offers dynamic thread summaries & writing coaching, user-level *Prioritize My Inbox* logic rules, and text drafting matching unique individual voices	It lacks sorting ground specifically in institutional data objects (e.g., academic standing, registration holds)	It offers AI-based academic communication classification, and ML-driven institutional prioritization
Automation & Analytics	It offers basic personal follow-up tracking & reminders, and Work IQ semantic organization model filters	It lacks automation loops triggered by live campus data (e.g., low attendance, deadline failures)	It offers academic event-triggered automation engine, campaign administration module, and institutional analytics & reporting

Table 11: Microsoft Outlook Gap

2.3 Analysis of Gaps and Limitations in Current Solutions

In this chapter, we've reviewed several systems that represent a large leap forward in new ways to communicate, work collaboratively and communicate via alert systems, automate workflows, and communicate using AI-based messaging systems. These systems are very strong in their individual components, however, when looking at the potential to build a better academic communication ecosystem from existing systems, it became apparent most current systems provide solutions to narrow areas of communication management and therefore do not support an integrated academic communication ecosystem. Therefore, educational providers are using multiple, non-integrated systems to manage notifications, messaging, campaigns, analytics, and academic communication workflows, leading to inefficiencies and administrative burden.

The literature review identified many key limitations in existing systems.

2.3.1 Lack of Centralized Academic Communication Management

The majority of communication platforms focus on general purpose collaboration, business communication or engagement with customers — certainly not intended as a mechanism for managing academic communication. The overwhelming majority provide you with the ability to messaging effectively; however, they do not integrate directly with your academic records, reporting systems or the institutional workflow.

The majority of email automation platforms, SMS communication applications all have their primary focus on delivering messages; however, the majority will require external interfaces in order to retrieve student records, attendance information, course data and information related to academic events.

As a result, the majority of higher education institutions end up using multiple, disconnected systems for supporting their communication process, which creates disconnected data and has a negative impact on staff productivity and work efficiency.

2.3.2 Limited Workflow Automation

Most of the systems in use are capable of sending messages and notifications but only support a very limited number of automated communications for creating, managing and executing a variety of workflows in an academic setting.

Automated communication is frequently required within educational institutions for a number of different situations that include:

- Attendance Shortage Warnings
- Deadlines associated with Assignments
- Exam Schedules
- Student Interventions and Retention Procedures
- Notification Chains for faculty members and students.
- Processes for Administrative Approvals
- Personalized Communication Campaigns

Some products such as Shortwave and other workflow automation tools provide limited amounts of automation; however, they were not designed specifically for educational institutions, and there are no methods available for the seamless integration of academic processes with these tools.

The proposed SACP (Smart Academic Communication Platform) would eliminate these limitations by providing a configurable rule-based workflow engine that will produce automated correspondence for an academic institution or user based upon academic data and events within an institution.

2.3.3 Absence of AI-based Classification and Prioritization System

Spark Mail, SaneBox, Shortwave, and Microsoft Outlook Copilot are all new tools used in combination with AI-powered technologies to provide users with additional features such as intelligent inboxes, summaries of emails, intelligent searches & priority types of emails.

These tools are designed primarily to enhance personal productivity & improve enterprise communications; while these tools are helpful for users in these contexts – they do not focus on helping users manage their communications for an educational institution. The way they determine which types of communications to prioritize, generally uses data surrounding: (Sender Behavior, User Interaction, Key Words, Conversations) without taking academic data into account.

Because of this limitation, users such as students, faculty & administration continue to have an increased number of notifications making it possible to miss: (Deadlines, Announcements, Events).

The proposed Student-Academic Communication Platform (SACP) will utilize machine learning models to assist users with:

- Identifying spam communications
- Identifying academic notifications to prioritize
- Intelligent message categorization
- Identifying high-risk communication scenarios
- Providing context-aware notifications using institutional data

These additional capabilities will help support the communication needs of educational institutions.

2.3.4 Weak Academic Analytics and Reporting

The majority of systems evaluated offer primary communication metrics: message delivery rate, open rate, click-through rate, and engagement statistics.

These metrics, although helpful, but do not provide enough data for evaluating institutional communication strategy effectiveness. Educational institutions need additional reporting capability, including:

- Analysis of overall communication campaign performance
- Monitoring of student engagement
- Visualization of notification delivery
- Measurement of academic communication effectiveness
- Reporting of all departmental communications
- Auditing of all institutional communications

Advanced analytical capabilities in the area of education analytics are critical to enabling the institution to make informed decisions relating to communication practices and engagement strategies with students.

2.3.5 Limited Academic Workflow Orchestration

Multiple education platforms, such as Canvas, Moodle, etc. have built-in notification systems and event-driven messaging capabilities.

However, many of these systems only offer a limited number of pre-defined communication scenarios with very few options available for creating automated workflows that are unique to each institution. More complex processes like attendance-based escalation chains, advisor intervention workflows, personalized student outreach campaigns, and context-rich academic alerts require extensive custom programming or the use of outside tools.

Additionally, most enterprise academic systems require administrative set-up and technical expertise to configure, making it very difficult for non-technical faculty or staff to access and use their systems.

The proposed SACP proposes configurable workflow orchestration that will combine automation rules, communications templates, reporting/analytics, and AI-assisted prioritization on a single, unified platform.

2.3.6 Fragmented Communication Ecosystems

A growing number of institutions have begun using two or more different solutions to communicate with their faculty members and students, through:

- Gmail
- WhatsApp
- Google Classroom
- Discord

This fragmented approach increases operational complexity, creates integration challenges and can lead to inconsistent communication experiences for both students and staff.

The primary goal of SACP is to consolidate these communication functions into a single, centralized, distributed platform to enable unified communication management.

2.3.7 Compliance, Privacy, and Institutional Governance Challenges

Institutions have a requirement to independently manage their compliance requirements associated with communication and privacy regulations, consent to communicate with students, and data governance across multiple communication platforms.

Although some systems (e.g., WhatsApp Business, TrueDialog, and enterprise email platforms) have developed security controls and consent mechanisms that provide for compliance, they lack university-based governance programs that meet the institution's communication policy objectives.

In addition, educational institutions need communication systems that can provide privacy protection, role-based access, auditing of communication, and responsible use of communications and other student information.

The SACP has been designed to create a centralized communication platform and include institutional governing mechanisms to facilitate communication and communication policy compliance.

2.3.8 Scalability and Integration Challenges

A number of reviewed systems have costly enterprise subscription requirements, are heavily reliant upon proprietary ecosystems, and offer limited customization options. Institutions often face problems integrating communication platforms with their current academic infrastructure, creating increased implementation complexity and operational costs.

The need for scalable, easily customizable, and modularly extensible communication solution has become critical to enable educational institutions to adapt quickly and easily to changing academic needs without being overly dependent on a third-party vendor.

The proposed SACP will eliminate these challenges by providing a modularly distributed architecture that allows for extensibility, customization at the institution level, and the ability to integrate with academic systems while at the same time providing a scalable platform for future growth.

Chapter 3

SYSTEM ANALYSIS

3.1 Introduction

In this chapter, we'll conduct a system analysis on our project including requirement gathering, functional and non-functional requirements, user roles, use-case diagrams, domain model diagram, and data flow diagrams.

We'll include all components of the project and explain them in the scope of system analysis with the help of proper diagrams.

3.2 Requirement Gathering

In this phase, we'll define, collect, analyze, and document prior requirements of the system being designed and built. This will ensure that the Smart Academic Communication Platform (SACP) meets the communications needs of educational institutions, while also meeting the needs of all types of user roles as per defined in user roles section.

Several requirement gathering techniques were utilized during the analysis phase of the project.

3.2.1 Requirement Gathering Techniques

3.2.1.1 Literature Review

The literature review examined many types and kinds of providers' systems such as communications and notification systems to identify and analyze the strengths, weaknesses and gaps in research with respect to these various systems including but not limited to: communication tools and platforms, workflow automation systems/tools, email campaigns platforms, learning management systems, student information systems/portal products, text messaging/paging providers, and AI based communications.

3.2.1.2 Comparative System Analysis

To identify the requirements for the proposed platform, we evaluated several systems that allow communication, such as Slack, Microsoft Teams, Discord, Rocket.Chat, Twilio, WhatsApp Business, etc. The analysis focused on evaluating:

- Basic features of each system
- Ability or capacity for automation
- Types of communication offered (i.e., text messaging, email, video conferencing)
- Types of integrated systems are currently supported by each system.

3.2.1.3 Academic Workflow Analysis

We conducted a survey for common communication processes within educational institutions to understand how students, faculty members, and administrators exchange information with each other. In the survey, attention was given to daily messages routine, drawbacks in current system, and get opinions for improvements.

3.2.1.4 Stakeholders Discussion

In discussion with the stakeholders including project team members, faculty members, and project supervisors, we further refined the requirements of our proposed project. These

helped in identifying challenges faced by stakeholders of academic institutions in daily communication workflow. And ensured that the proposed system remains aligned with institutional requirements.

3.2.2 Stakeholders Identification

The proposed Smart Academic Communication Platform involves multiple stakeholders who interact with the system in different capacities.

3.2.2.1 Students

Students are the primary recipients of academic communications. With the help of proposed system, they'd receive important notifications of announcements, reminders, alerts, and personalized communications related to academic activities.

3.2.2.2 Faculty Members

Faculty members are major users of the system because they need to send a lot of messages based on requirements. They'd utilize the platform to communicate effectively with students by creating templates, campaigns, and automated workflows.

3.2.2.3 Administrators

Administrators need to monitor and organize communication between faculty members and students so that everyone can get timely notification and don't miss important reminders and events. They can also manage academic data to make communication workflow better and more effective.

3.2.3 Existing System Analysis

Research into existing systems has shown that many communication platforms serve one or two functions, rather than an all-inclusive academic communications platform.

While collaboration platforms like Slack or Microsoft Teams allow for immediate messaging, they offer no integration into academic workflow systems. Notification services such as Twilio or WhatsApp Business only provide the means to deliver messages, requiring third-party systems to automate the workflow. Learning Management Systems have educational capabilities but provide little to no ability to manage communication. In addition, AI-based communication tools improve individual productivity but fail to solve communication problems for institutions.

As a result, these observations indicate a significant need for a single, centralized platform to combine communication channels, automate workflows, provide analytics, and manage intelligent notifications.

3.2.4 Problem Identification

Upon completing the requirement analysis of educational institutions, the following communication challenges have been identified;

- Multiple communication platforms used unnecessarily create fragmented communications.

- No centralized management of all forms of communications across the educational institution.
- Limited automation of academic communications workflows, creating inefficiencies in the way academic staff send out communications.
- Difficulty in measuring the effectiveness of communications made within the institution.
- Excessive manual efforts involved sending notifications.
- Overload of notifications and poor prioritization of received messages.
- Limited analytical insight regarding the performance of communications created and sent out.
- Limited integration of communication systems with academic records of students.

These challenges will ultimately affect the objectives and scope of the Smart Academic Communication Platform.

3.3 Functional Requirements

The Smart Academic Communication Platform has different functional features for each user role like admin, faculty member, and student. Each user can login into their own portal using the login system as well as register their account to get access based on the role.

Below will list all the functional requirements of this project:

3.3.1 FR01: Sign Up Form

Req. No	FR_01
FR01-01	The system will enable users to sign-up/register their account and authenticate based on their roles.
FR01-02	The system will redirect them on login form after successful registration.
FR01-03	The system will enable admin to add registry of faculty members and students first and then they will able to sign up with their access code or registration number, respectively.

Table 12: FR01 Sign Up

3.3.2 FR02: Login and Logout Form

Req. No	FR_02
FR02-01	The system will enable users to login to their registered account via their email and password.
FR02-02	The system will redirect admin/faculty member/student to their dashboard on successful login.
FR02-03	The system will also allow users to log out from their account and redirect them on landing page.

Table 13: FR02 Login and Logout Form

3.3.3 FR03: Reset Password & OTP

Req. No	FR_03
FR03-01	The system will allow users (admin/faculty members/students) to reset their passwords if they forget it.
FR03-02	The system will send an OTP to the user's email with a time limit of expiry.
FR03-03	The system will redirect the user to reset password screen on correct OTP entry.
FR03-04	The system will notify the user on successful password reset and redirect him/her on the login form.

Table 14: FR03 Reset Password & OTP

3.3.4 FR04: Landing Page

Req. No	FR_04
FR04-01	The system will show landing page to every user who has not logged in or wants to visit our platform.

Table 15: FR04 Landing Page

3.3.5 FR05: Dashboard

Req. No	FR_05
FR05-01	The system will enable users to access their dashboard portal according to their role (admin/faculty members/student).
FR05-02	The system will display specific analytical data to users based on their roles such as communication engagement analytics, total number of students and faculty, attendance analytics, average response time, attendance analytics, etc.
FR05-03	All kinds of data will be shown to admin, but faculty members can only show data specific to their department and students can only show data depending on their specific requirements such as attendance, CGPA, etc.

Table 16: FR05 Dashboard

3.3.6 FR06: Academic Data Module

Req. No	FR_06
FR06-01	The system will allow admin and faculty members to insert, delete, update, and remove academic data with limited access based on their role.
FR06-02	The admin can create foundational records for academic data and read/update/remove them. He can only read records generated by faculty members and students.
FR06-03	A faculty member can create academic records related to his/her assigned courses, students whom he/her teach, and exams conducted by him/her. Although, he/her can also read, update and remove them.
FR06-04	A student can only read all academic records related to him/her and send his/her queries to faculty members/admin.

Table 17: FR06 Academic Data Module

3.3.7 FR07: Template Management Module

Req. No	FR_07
FR07-01	The system will allow admin and faculty members to create, delete, update, and remove templates for their specific use cases in campaigns.
FR07-02	Admin can create templates and use them in campaigns for faculty members and students as selected target audience.
FR07-03	Faculty members can create templates and use them in campaigns for students as selected target audience.
FR07-04	Students can only receive messages generated by templates by admin and faculty members.
FR07-05	Templates use placeholders for dynamic data insertion.
FR07-06	Admin has access to all placeholders such as department, faculty members, students, courses, etc.
FR07-07	Faculty members have access to specific placeholders based on their department, courses, students, etc.
FR07-08	Templates will allow admin and faculty members to send personalized messages which help to increase attention of target audience.
FR07-09	As students can't create templates, there is no need for placeholders' access.

Table 18: FR07 Template Management Module

3.3.8 FR08: Campaign Management Module

Req. No	FR_08
FR08-01	The system will allow admin and faculty members to create, delete, update, and remove campaigns for their specific needs.
FR08-02	Campaigns will allow admin and faculty members to minimize their efforts on sending messages to large audiences by reducing human efforts and automated messages workflows.
FR08-03	Campaigns can use templates for the messages to reduce human effort seamlessly.
FR08-04	Campaigns will use event-based and rule-based triggering to make them automated and repeated without much human (admin or faculty member) interaction.
FR08-05	Students can only receive messages/notifications from these campaigns and post their queries.

Table 19: FR08 Campaign Management Module

3.3.9 FR09: Automation Rule Engine

Req. No	FR_09
FR09-01	Automation rule engine is integrated with campaigns which allows them to send messages to target audience when a specific rule or event is triggered.
FR09-02	This feature is limited to only admin and faculty members, because they are the ones who can create campaigns.
FR09-03	This feature helps in reducing human effort and provides a platform where admin and faculty members can create automated messages workflow.

Table 20: FR09 Automation Rule Engine

3.3.10 FR10: Messaging Service

Req. No	FR_10
FR10-01	The system will allow all users to send and receive messages via email and SMS.
FR10-02	This will be used in campaigns, queries, and real time communication system.
FR10-03	A message queue system will be integrated to handle failed messages and resend them later.

Table 21: FR10 Messaging Service

3.3.11 FR11: Notification System

Req. No	FR_11
FR11-01	The system will allow messaging services to send notifications to target audience.
FR11-02	This feature has some additional functionality such as reminder on important unseen messages.

Table 22: FR11 Notification System

3.3.12 FR12: Real-time Communication Module

Req. No	FR_12
FR12-01	The system will integrate a real-time (one-to-one) communication module between faculty members and students.
FR12-02	This system will allow faculty members and students to do discussions on campaigns (messages) privately.
FR12-03	This will help faculty members to organize their chat and help students to clear their doubts based on their queries.
FR12-04	The faculty member can also send a general message to entire/selected students of the chat to save time and effort by avoiding manual effort of resending same message repeatedly on each channel.

Table 23: FR12 Real Time Communication Module

3.3.13 FR13: Message Classification and Prioritization System

Req. No	FR_13
FR13-01	The system will allow AI/ML models to classify and prioritize messages sent by admin, faculty members, and students.
FR13-02	When a student sends a query the system will use the ML model to prioritize it into HIGH, MEDIUM, or LOW priority. This will help faculty members to reply to important queries first and then move towards lower priority queries. This can save a lot of time and effort which could be wasted on focusing on useless and less important queries.
FR13-03	The system will detect spam messages and move them immediately into separate "spam" box from inbox to prevent interruption on faculty members' and admin's work.

Table 24: FR13 Message Classification and Prioritization System

3.3.14 FR14: Attachments Support with Messages

Req. No	FR_14
FR14-01	The system will enable members to attach document files with their messages to support document sharing among them.
FR14-02	This feature will be integrated with campaigns and real time communication module.
FR14-03	Students can only attach files in real time communication with a limited file size.
FR14-04	The system will also scan all attached files for viruses to prevent cyber theft and data leaks.

Table 25: FR14 Attachment Support

3.3.15 FR15: Analytics and Reporting

Req. No	FR_15
FR15-01	The system will enable users to see analytics and reporting charts on their dashboards.
FR15-02	Admin can see overall general and average analytics of academic data and campaigns engagement.
FR15-03	Faculty members can see overall analytics of their (specific) academic data and campaigns engagement. Also, they can see analytics of unseen messages.
FR15-04	Students can see analytics of their academic performance and average response time of faculty members and admin on their queries.

Table 26: FR15 Analytics and Reporting

3.4 Non-Functional Requirements

The Smart Academic Communication Platform is not only focused on what the system does but also on how well it does things. These are called non-functional requirements. They make sure the system works smoothly, is safe to use, and gives the users good experience.

3.4.1 Performance

The system is designed to be fast and responsive so that all users can work without delays. It should handle many users at the same time without slowing down the server. Also, when processing large data or running background jobs, the system should give quick and correct results, even if there is a lot of data to manage.

3.4.2 Security

The system makes sure that all users log in securely using a strong authentication process. It uses role-based access control so that each user can only see and use the features meant for their role. Also, sensitive information like passwords is kept safe by encrypting the data both when it is sent over the internet and when it is stored. It also scans all attached files to messages for viruses to protect the system from theft and data leaks.

3.4.3 Usability

The system is built to stay online and available almost all the time, aiming for 99.9% uptime. If any errors happen, the system handles them smoothly by showing clear and helpful messages to users without confusing technical details.

The system will provide user-friendly interfaces to give smooth experience while interacting with the system and perform their work seamlessly.

3.4.4 Maintainability

The system's code is written carefully following clean coding standards and vertical slice architecture, which means it is easy to read, understand and maintain. Good documentation is also provided, so anyone working on this project in the future can quickly learn how it works and make changes. The system is designed in a modular way, meaning it is divided into smaller parts or components. This makes it possible to update or replace one part without causing problems in other areas. To keep the system reliable, automated tests are set up to regularly check if everything is working correctly. These tests help catch errors early, especially when new changes or updates are made.

3.5 GANTT Chart

3.6 Use Case Diagrams

Chapter 4

SYSTEM DESIGN

4.1 System Architecture

An architecture diagram is a simple picture that shows how different parts of a system are connected and work together. This project uses a web based three-layer architecture system.

- The first layer is the Frontend, where users will see and use the website.
- The second layer is the Backend, where all logic happens.
- The third layer is the Database, where all the data is safely stored.

Diagram will be inserted soon...

4.2 Database Design

The Smart Academic Communication Platform (SACP) utilizes a relational database that will store and manage data for multiple connected entities. To organize all the information, it collects data and uses a database schema to allow for: user management; academic records; communications workflow; managing campaigns; managing templates; sending notifications; and performing analytics. Each entity has its own separate table with associated attributes and relationships to provide data consistency, integrity and optimum retrieval. The following subsections give a detailed description of each table and attributes.

4.2.1 User Table

Description: This table stores user details like username, email, password, and role in the system.

Field Name	Data Type	Constraint	Description
id	UUID	primary key (pk)	Unique identifier of user
avatar	bytea	nullable	Profile picture of user
username	varchar	required	Username of user
email	varchar	unique	Email of user
password_hash	string	required	Encrypted password of user
role	enum	required	Role of user (Admin/Faculty Member/Student)
is_active	boolean	required	Shows if user is active or not. Inactive users can't access their account.
created_at	datetime	required	Shows when user is created

Table 27: User Table